

Python Assignment 15

1. Write a lambda function using map() which accepts a list of numbers and returns a list of squares of each number.

Answer :

```
Square = lambda Nlist : (Nlist ** 2)

def main():
    print("Enter the list of Numbers : ")
    NumList = list(map(int,input().split()))

    Mdata = list(map(Square, NumList))
    print(Mdata)

if __name__ == "__main__":
    main()
```

2. Write a lambda function using filter() which accepts a list of numbers and returns a list of even numbers.

Answer :

```
def main():
    print("Enter the list of Numbers : ")
    NumList = list(map(int, input().split()))
    print("You Entered List is : ", NumList)

    Even = filter(lambda iNumlist:iNumlist%2==0,NumList)
    print("Even Numbers : ",(Even))

if __name__ == "__main__":
    main()
```

3. Write a lambda function using filter() which accepts a list of numbers and returns a list of odd numbers.

Answer :

```
def main():
    print("Enter the list of Numbers : ")
```

```

NumList = list(map(int,input().split()))

OddNumbers = filter(lambda iNumList : iNumList%2!=0,NumList)
print("Odd Numbers : ",list(OddNumbers))

if __name__ == "__main__":
    main()

```

4. Write a lambda function using reduce() which accepts a list of numbers and returns the addition of all elements.

Answer :

```

from functools import reduce

def Addition(iNo1,iNo2):
    return iNo1 + iNo2

def main():
    print("Enter the list of Numbers : ")
    NumList = list(map(int,input().split()))

    Result = reduce(Addition, NumList)
    print("Your All Numbers Addition is : ", Result)

if __name__ == "__main__":
    main()

```

5. Write a lambda function using reduce() which accepts a list of numbers and returns the maximum element.

Answer :

```

from functools import reduce

def BigBull(iNo1,iNo2):
    if iNo1 > iNo2:
        return iNo1
    else:
        return iNo2

def main():

```

```

print("Enter the list of Numbers : ")
NumList = list(map(int,input().split()))

Result = reduce(BigBull, NumList)
print("Your Big Number is : ", Result)

if __name__ == "__main__":
    main()

```

6. Write a lambda function using reduce() which accepts a list of numbers and returns the minimum element.

Answer :

```

from functools import reduce

def ChotaDon(iNo1,iNo2):
    if iNo1 > iNo2:
        return iNo2
    else:
        return iNo1

def main():
    print("Enter the list of Numbers : ")
    NumList = list(map(int,input().split()))

    Result = reduce(ChotaDon, NumList)
    print("Your Small Number is : ", Result)

if __name__ == "__main__":
    main()

```

7. Write a lambda function using filter() which accepts a list of strings and returns a list of strings having length greater than 5.

Answer :

```

Digits = lambda iNumList : len(iNumList) > 5

def main():
    print("Enter the list of Numbers : ")
    NumList = list(map(str,input().split()))

```

```

    FilterList = list(filter(Digits, NumList))
    print(FilterList)

if __name__ == "__main__":
    main()

```

8. Write a lambda function using filter() which accepts a list of numbers and returns a list of numbers divisible by both 3 and 5.

Answer :

```

Divisible = lambda iNumList : iNumList%3==0 and iNumList%5==0

def main():
    print("Enter the list of Numbers : ")
    NumList = list(map(int,input().split()))

    DivisibleList = list(filter(Divisible, NumList))
    print(DivisibleList)

if __name__ == "__main__":
    main()

```

9. Write a lambda function using reduce() which accepts a list of numbers and returns the product of all elements.

Answer :

```

from functools import reduce

def main():
    print("Enter the list of Numbers : ")
    NumList = list(map(int,input().split()))

    Products = reduce(lambda iNum1, iNum2 : iNum1 * iNum2, NumList)
    print("Product of all Elements : ", Products)

if __name__ == "__main__":
    main()

```

10. Write a lambda function using filter() which accepts a list of numbers and returns the count of even numbers.

Answer:

```
from functools import reduce

def main():
    print("Enter the list of Numbers : ")
    NumList = list(map(int, input().split()))    #map

    FilterList = list(filter(lambda iNumList : iNumList%2 ==
0, NumList))    #filter
    print("This is Even Numbers in your List : ", FilterList)

    ReduceNumber = reduce(lambda iNum1, iNum2 : iNum1 + iNum2 , FilterList)
    print("Addition of even Numbers : ", ReduceNumber)
    print("Count of filter List is : " , len(FilterList))

if __name__ == "__main__":
    main()
```